Digital Hospital Appointment & Patient Management System

Business Requirement Document (BRD)

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1. Introduction

This Business Requirements Document (BRD) outlines the business requirements for the Hospital Appointment & Patient Management System. The purpose of this document is to define the business objectives, project scope, stakeholder requirements, and functional capabilities needed to streamline patient registration, appointment scheduling, medical record management, billing, and related hospital operations.

This document serves as a reference for stakeholders, business analysts, developers, testers, and project managers throughout the project lifecycle and helps ensure alignment between business needs and system functionality.

2. Project Overview

Project Name: Digital Hospital Appointment & Patient Management System (DHAPMS)

Industry: Healthcare / Hospital

Client: City Care Multi-Specialty Hospital (10 branches)

Project Duration: 8 months

Methodology: Agile (Scrum)

- **Role:** Senior Business Analyst
- **Responsibilities:**
  - Requirement Gathering and Analysis
  - BRD Preparation
  - Stakeholder Management
  - User Story Creation
  - Process Flow Modelling
  - UAT Planning and Coordination

3. Business Problems

The hospital relies on manual and disconnected processes for patient registration, appointment scheduling, medical record management, billing, and insurance handling. This leads to operational inefficiencies, increased administrative workload, scheduling conflicts, data errors, longer patient wait times, and a poor patient experience.

4. Business Objective

Operational Efficiency — Automate scheduling and registration to reduce manual workload and errors.

Patient Experience — Enable easy appointment booking with reminders to minimize wait times.

Centralized Records — Maintain unified patient data accessible to authorized staff in real time.

Resource Optimization — Maximize utilization of doctors, rooms, and equipment through smart scheduling.

Billing & Revenue — Streamline payments and insurance claims to reduce revenue leakage.

Compliance & Security — Ensure patient data privacy per healthcare regulations (HIPAA/local standards).

Reporting & Insights — Provide dashboards for patient flow, staff performance, and operational trends.

Scalability — Support integration with labs, pharmacy, and insurance systems as the hospital grows.

To improve patient care and operational efficiency by automating appointment booking, patient record management, and hospital administrative processes.

5. Scope

The Hospital Appointment & Patient Management System is a web-based platform designed to streamline and automate key hospital operations from patient registration through discharge. The system centralizes patient information, appointment scheduling, billing, and administrative processes to improve operational efficiency, reduce manual effort, and enhance the overall patient experience.

In Scope

- Patient registration and profile management
- Electronic patient record management
- Appointment booking, rescheduling, and cancellation
- Doctor availability and schedule management
- Billing and payment processing
- Insurance claim management
- Laboratory report management
- Prescription and pharmacy record management
- Discharge summary generation
- Automated notifications and appointment reminders
- Role-based access for patients, doctors, receptionists, and administrators
- Dashboard and reporting for operational monitoring

Out of Scope

- Telemedicine and video consultations
- AI-based diagnosis or predictive analytics
- IoT and wearable device integration
- Mobile application development
- Integration with government healthcare databases
- Advanced inventory and supply chain management
- Third-party hospital network integration

- Multi-hospital management functionality

This project focuses on digitizing core hospital workflows to improve service delivery, data accuracy, and operational effectiveness.

6. Stakeholder Analysis

Primary Stakeholder

Stakeholder	Interest	Influence	Role in System
Patient	High	Medium	Books appointments, views medical records, receives reports, and makes payments
Doctor	High	High	Reviews patient history, conducts consultations, updates diagnoses, and prescribes treatments.
Receptionist	High	Medium	Registers patients, manages appointments, and handles front-desk operations.
Hospital Administrator	High	High	Oversees hospital operations, user management, and reporting.
Pharmacist	Medium	Medium	Manages prescriptions and medicine dispensing records.
Lab Technician	Medium	Medium	Conducts tests and uploads laboratory reports.
Billing Executive	High	Medium	Generates invoices, processes payments, and manages billing activities.
Insurance Coordinator	High	Medium	Handles insurance verification and claim processing.

Secondary Stakeholder

Stakeholder	Interest	Influence	Role in System
Nurse	Medium	Low	Assists doctors and updates patient care information.

Department Head	High	Medium	Monitors departmental performance and service quality.
Finance Team	Medium	High	Tracks revenue, payments, and financial reporting.
IT Support Team	High	High	Maintains system availability and technical support.
System Administrator	High	High	Manages user access, roles, permissions, and configurations.
Compliance Team	Medium	High	Ensures adherence to healthcare regulations and audit requirements.
Hospital Management	High	High	Reviews KPIs and supports strategic decision-making.

External Stakeholder

Stakeholder	Interest	Influence	Role
Insurance Company	Medium	Medium	Claim approval/rejection
Payment Gateway Provider	Low	Medium	Online payment processing
SMS/Email Notification Vendor	Low	Low	OTP and alerts
Third-party Lab Systems	Low	Medium	Lab integrations
Government Healthcare Authority	Low	High	Compliance/reporting
Pharmacy Suppliers	Low	Low	Medicine inventory supply

The primary stakeholders are patients, doctors, receptionists, hospital administrators, pharmacists, laboratory technicians, billing executives, and insurance coordinators, as they directly interact with the system. Secondary stakeholders include nurses, finance teams, IT support teams, compliance teams, and hospital management. External stakeholders such as insurance companies, payment gateway providers, and regulatory authorities support business operations and system integration.

7. Requirement Gathering Approach

1. Elicitation Techniques Used

Technique	How You Used It
Interviews	Conducted one-on-one sessions with doctors, receptionists, and finance team to understand pain points
Workshops	Organized group sessions with all stakeholders to align on priorities and resolve conflicts
Observation	Shadowed receptionists for 2 days to understand current manual appointment process
Document Analysis	Reviewed existing appointment registers, billing formats, and patient record files
Prototyping	Shared wireframes with stakeholders to visually validate requirements before documentation
Survey	Collected patient feedback through structured questionnaires to understand patient pain points

2. Requirement Gathering Process

Requirement gathering was conducted over a 3-week period. The process began with stakeholder identification and a kickoff meeting to align on project goals. One-on-one interviews were conducted with each department to capture individual requirements. Group workshops were held to resolve conflicting requirements and finalize priorities using the MoSCoW technique. Observations were conducted at the reception desk to understand the current AS-IS process and identify gaps. All gathered requirements were documented in the BRD, reviewed with stakeholders, and formally signed off before development began. A Requirements Traceability Matrix (RTM) was maintained throughout the project to ensure every business requirement was mapped to its corresponding user story, test case, and UAT scenario.

3. Output of Requirement Gathering

Output	Document
Business Requirements	BRD
Process Flows	AS-IS and TO-BE diagrams
Screen Mockups	Wireframes in Figma
User Stories	JIRA backlog
Traceability	RTM in Excel
Use Case Diagrams	Lucidchart / Draw.io
Meeting Notes & Sign-Off	Microsoft Word / Email

8. Assumptions & Constraints

Assumptions

1. The hospital operates as a multi-department facility with both Outpatient (OPD) and Inpatient (IPD) services requiring appointment scheduling.
2. Patients possess or can create a unique digital identity (mobile number/email) for account registration; a minority without digital access will be supported via front-desk-assisted booking.
3. Doctors/consultants provide their availability schedules in advance, which are configurable by the hospital administration.
4. The hospital already has an internal network and basic IT infrastructure capable of supporting a cloud or on-premise deployment.
5. Payment gateway and SMS/email notification services are available as third-party integrations and do not require in-house development.
6. Patient medical records created within the system are assumed to be the primary digital source of truth going forward, though legacy paper records may coexist during transition.
7. Hospital staff (doctors, nurses, front-desk, lab technicians) will receive role-based training and are assumed willing to adopt the new system following change management efforts.
8. Regulatory requirements for health data protection (e.g., DPDP Act 2023) are assumed applicable and will guide data handling design.
9. Diagnostic and lab systems are assumed to support standard integration protocols (HL7/FHIR) or will be upgraded to do so.
10. The initial rollout is assumed to be for a single hospital/branch, with multi-branch scalability considered a future phase.

Constraints

1. **Budget Constraint** — Development and infrastructure costs are limited to the sanctioned project budget; premium third-party integrations (advanced AI diagnostics, telemedicine video SDKs) may be deferred to later phases.
2. **Timeline Constraint** — The system must be delivered within the defined project timeline, which may limit the scope of features in the initial release (MVP-first approach).
3. **Regulatory Constraint** — The system must comply with applicable healthcare data protection and medical record retention laws, which govern how patient data is stored, accessed, and shared.
4. **Technology Constraint** — Integration is limited to systems that support standard interoperability protocols (HL7/FHIR); legacy systems without API support may require manual workarounds or middleware.
5. **Infrastructure Constraint** — Reliable internet connectivity is required for real-time appointment booking and EHR access; system performance in low-bandwidth hospital zones may be a limiting factor.
6. **Staffing Constraint** — Availability of trained IT and clinical staff for UAT, training, and post-launch support is limited to existing hospital resources.
7. **Data Migration Constraint** — Only structured legacy data (in digital format) can be migrated within scope; scanned/paper records require manual digitization, which is a separate, resource-intensive effort.
8. **Device Constraint** — The patient-facing application must function on commonly available low-to-mid-range smartphones, limiting the use of high-end UI/graphics features.
9. **Language Constraint** — Initial release supports English (and optionally one regional language); full multilingual support is out of scope for MVP.
10. **Third-Party Dependency Constraint** — Features reliant on external vendors (payment gateway, SMS gateway) are constrained by those vendors' SLAs, uptime, and API rate limits.

9. Risk

Risk ID	Risk Description	Category	Probability	Impact	Mitigation Strategy	Owner
R-01	Continuous addition of	Business	High	Medium	Enforce formal change request	Business Analyst

	new feature requests during development leads to scope creep and delays				(CR) process tied to a baselined BRD; route all changes through impact assessment and sponsor approval	
R-02	Clinical and administrative staff resist transitioning from manual/paper-based workflows to the digital system	Operational	High	Medium	Conduct structured training programs, phased rollout, and appoint department-level change champions	Hospital Administrator
R-03	Legacy patient records are lost, duplicated, or corrupted during migration to the new EMR	Operational	Medium	High	Execute phased migration with validation checkpoints, parallel-run period, and a documented rollback plan	Project Manager / BA
R-04	Integration failures between the platform and existing lab, pharmacy, or insurance/TPA systems	Technical	Medium	High	Adopt HL7/FHIR interoperability standards; conduct early interface testing with all third-party systems	Technical Lead
R-05	Patient health records (PHI/PII) exposed due to inadequate access controls, weak	Security	Medium	High	Implement role-based access control (RBAC), end-to-end encryption, multi-factor authentication,	System Administrator

	encryption, or unauthorized access				and periodic security audits	
R-06	System fails to comply with healthcare data protection regulations (e.g., DPDP Act 2023, applicable health data laws)	Compliance	Low	High	Engage legal/compliance SMEs during requirements phase; build configurable consent management and audit-trail modules	Compliance Officer / BA
R-07	Third-party services (payment gateway, SMS/email notification provider, insurance API) experience downtime or failure	Technical	Medium	Medium	Define SLAs with vendors; maintain fallback/alternate providers for critical services	Technical Lead
R-08	Project milestones are delayed due to underestimated effort, dependency bottlenecks, or approval delays	Schedule	Medium	Medium	Maintain a realistic, buffer-inclusive project schedule; track progress via regular sprint/milestone reviews	Project Manager
R-09	Key project resources (BA, developers, clinical SMEs) become unavailable during critical phases	Resource	Medium	Medium	Cross-train team members; maintain a resource backup plan and early escalation	Project Manager

					process for critical roles	
R-10	Inaccurate, incomplete, or duplicate patient data entered during registration affects downstream modules (EMR, billing, claims)	Business	Medium	High	Implement unique patient ID (UHID) generation with de-duplication logic and mandatory field validation at registration	Business Analyst
R-11	Appointment slot conflicts or double-booking due to concurrent booking requests	Operational	High	Medium	Implement real-time slot locking and concurrency control at the booking module level	Technical Lead
R-12	Delay or failure in critical notification delivery (appointment reminders, lab alerts) due to gateway or configuration issues	Operational	Low	Medium	Implement delivery status tracking with automatic retry and front-desk fallback for failed notifications	Technical Lead
R-13	Reporting dashboard reflects inaccurate data due to reconciliation gaps between source modules (billing, EMR, claims)	Technical	Low	Medium	Establish automated data reconciliation checks between source systems and the reporting layer	Technical Lead

10.Business Rule

Module 1: Patient Registration

1. Every patient must be assigned a unique Hospital ID (UHID) upon first registration; this ID persists across all future visits.
2. Registration requires mandatory fields: full name, date of birth, gender, contact number, and address.
3. Duplicate patient records must be flagged and merged if the same name, date of birth, and mobile number combination already exists.
4. Minors (under 18) must be registered with a linked guardian/parent contact for consent and communication.
5. Patients can self-register via the app/web portal, or be registered by front-desk staff for walk-ins.

Module 2: Appointment Booking

6. Patients can book, reschedule, or cancel an appointment only up to a defined cutoff (e.g., 2 hours before slot time).
7. Each appointment slot can be booked by only one patient at a time; concurrent booking attempts must trigger real-time slot locking.
8. Appointment booking requires selection of department, doctor (or "any available"), and preferred time slot.
9. Walk-in patients are queued separately from pre-booked patients but follow the same doctor's available time window.
10. A patient cannot hold more than 2 active (unfulfilled) appointments with the same doctor at any given time.
11. Emergency/urgent cases bypass standard slot booking and are routed through a priority queue.

Module 3: Doctor/Provider Scheduling

12. Doctors must publish their weekly availability at least 7 days in advance; changes within 48 hours require admin approval.
13. A doctor's leave or unavailability automatically blocks new bookings and triggers rescheduling prompts for existing appointments.
14. Maximum patients per time slot per doctor is configurable by department (default: 1 patient per 15-minute slot).
15. Doctors can view only their own schedule and patient list; cross-doctor visibility requires admin/department-head role.

Module 4: Electronic Health Records (EHR)

16. Each patient's medical history, prescriptions, and diagnostic reports are stored under their unique UHID and are cumulative across visits.
17. Only authorized clinical staff (treating doctor, assigned nurse) can edit a patient's active EHR entry; all edits are logged with timestamp and user ID.
18. Patients have read-only access to their own EHR summary (diagnosis, prescriptions, reports) via the patient portal.
19. EHR data must not be permanently deleted; corrections are handled via versioned amendments, preserving the original entry.

Module 5: Billing & Payments

20. All billable services (consultation, diagnostics, procedures) must be linked to a valid appointment or admission record before invoicing.
21. Payments can be made via cash, card, UPI, or insurance/TPA settlement; partial payments are allowed with balance tracking.
22. An invoice is auto-generated upon consultation completion or service delivery and cannot be edited post-generation (only credit notes can offset errors).
23. Insurance-covered patients require pre-authorization validation before cashless billing is applied.

Module 6: Lab & Diagnostics

24. Lab test orders must be linked to a doctor's prescription; walk-in lab tests without a prescription require admin override with justification.
25. Diagnostic reports are uploaded against the patient's UHID and automatically linked to the ordering doctor's dashboard.
26. Critical/abnormal lab results must trigger an immediate alert to the ordering doctor, separate from standard report delivery.

Module 7: Notifications

27. Patients receive automated SMS/email/app notifications for appointment confirmation, reminders (24 hours and 2 hours prior), and cancellations.
28. Doctors receive notifications for schedule changes, new bookings, and cancellations affecting their calendar.
29. Failed notification delivery (invalid number/email) must be logged and flagged for front-desk follow-up.

Module 8: Cancellation & Rescheduling

30. Cancellations made after the cutoff window are recorded as "late cancellations" and may be subject to a no-show/cancellation policy (if configured).
31. A cancelled slot is immediately released back into the available pool for other patients to book.
32. Three consecutive no-shows by a patient trigger a review flag on their profile, requiring front-desk confirmation for future bookings.

Module 9: Admin Portal

33. Only Admin and Super Admin roles can create, modify, or deactivate doctor/staff accounts.
34. Department and doctor availability templates can only be configured by Admin or Department Head roles.
35. All administrative actions (user creation, role changes, schedule overrides) are logged in an audit trail with user ID and timestamp.

Module 10: Reporting & Compliance

36. Patient data exports (for referrals, insurance, legal requests) must be logged with requester identity and purpose.
37. All financial and clinical reports must reconcile against the source transaction/EHR data — no manually entered report figures are permitted.

11.User Roles & Permission

Purpose

The user roles within the Hospital Appointment & Patient Management System and the business-level permissions assigned to each, following the principle of **least privilege** and **Role-Based Access Control (RBAC)**. Each role is granted only the access necessary to perform its designated responsibilities, minimizing risk to patient data, financial records, and system integrity.

Permission Legend

Code	Permission	Description

V	View	Read-only access to information
C	Create	Ability to add new records
U	Update	Ability to modify existing records
D	Delete	Ability to remove/deactivate records
A	Approve	Authority to validate, sign off, or authorize an action
M	Manage	Full administrative control over a module (includes configuration)
-	No Access	Module not accessible to this role

Role-by-Role Detail

1. Patient

- **Responsibilities:** Register and maintain own profile; book, reschedule, or cancel appointments; view own medical history, lab reports, and prescriptions; make payments and submit insurance claim documents.
- **Key Permissions:** Create/Update own profile; View own EMR (read-only); Create/Update/Delete own appointments; View/Create own billing and insurance records.
- **Modules Accessible:** Patient Registration, Patient Records (EMR – own only), Appointment Management, Doctor Schedule (view only), Laboratory (own reports), Pharmacy (own prescriptions), Billing & Payments (own), Insurance Claims (own).

2. Doctor

- **Responsibilities:** Review assigned patient records; conduct consultations; order lab tests and prescribe medication; manage personal availability; certify insurance claims requiring medical justification.
- **Key Permissions:** View/Create/Update EMR for assigned patients only; Update own schedule; Create lab orders and prescriptions; Approve insurance claims requiring clinical sign-off.
- **Modules Accessible:** Patient Registration (view), Patient Records (EMR), Appointment Management, Doctor Schedule Management, Laboratory, Pharmacy, Billing & Payments (view), Insurance Claims, Reports & Dashboards (own patients).

3. Receptionist

- **Responsibilities:** Register walk-in and returning patients; manage appointment bookings across all doctors; initiate billing and insurance intake; provide front-desk support.

- **Key Permissions:** Create/Update patient demographic records; Full booking management (Create/Update/Delete); Initiate billing and insurance entries.
- **Modules Accessible:** Patient Registration, Patient Records (demographic view only), Appointment Management, Doctor Schedule (view), Laboratory (status view), Billing & Payments (initiate), Insurance Claims (initiate), Reports & Dashboards (front-desk).

4. Nurse

- **Responsibilities:** Record patient vitals and clinical notes for assigned patients; assist doctors during consultations; initiate lab orders per clinical protocol; support ward-level care coordination.
- **Key Permissions:** Create/Update EMR entries (vitals/notes) for assigned patients; Create lab orders per protocol; Update assigned appointment status.
- **Modules Accessible:** Patient Registration (view), Patient Records (EMR – assigned), Appointment Management (assigned), Doctor Schedule (view), Laboratory, Pharmacy (view), Reports & Dashboards (ward-level).

5. Lab Technician

- **Responsibilities:** Process lab test orders; upload and manage diagnostic results; maintain laboratory records and equipment logs; flag critical results for doctor attention.
- **Key Permissions:** Full management of laboratory test records (Create/Update/Manage); View patient information relevant to ordered tests only.
- **Modules Accessible:** Patient Registration (view), Patient Records (test-relevant view), Appointment Management (lab visits), Laboratory, Billing & Payments (test charges view), Reports & Dashboards (lab reports).

6. Pharmacist

- **Responsibilities:** Dispense prescribed medication; manage pharmacy inventory; verify prescriptions against patient records; process pharmacy-related billing.
- **Key Permissions:** Full management of pharmacy module (Create/Update/Manage); View prescription-relevant patient data; Create pharmacy billing entries.
- **Modules Accessible:** Patient Registration (view), Patient Records (prescription-relevant view), Pharmacy, Billing & Payments (pharmacy), Reports & Dashboards (pharmacy reports).

7. Billing Executive

- **Responsibilities:** Generate and manage patient invoices; process payments across departments; reconcile billing with insurance claims; maintain financial records.
- **Key Permissions:** Full management of billing module (Create/Update/Manage); Create/Update insurance claim entries; View charges from laboratory and pharmacy modules.
- **Modules Accessible:** Patient Registration (view), Appointment Management (view), Laboratory (charges), Pharmacy (charges), Billing & Payments, Insurance Claims, Reports & Dashboards (financial).

8. Insurance Coordinator

- **Responsibilities:** Process and validate insurance claims; liaise with TPAs/insurers for pre-authorization and settlement; ensure claims comply with policy terms; manage the end-to-end claims lifecycle.
- **Key Permissions:** Full management and approval authority over insurance claims (Create/Update/Approve/Manage); View relevant patient, billing, and clinical data needed for claim validation.
- **Modules Accessible:** Patient Registration (view), Patient Records (claim-relevant view), Appointment Management (view), Laboratory (view), Pharmacy (view), Billing & Payments (view), Insurance Claims, Reports & Dashboards (claims).

9. Hospital Administrator

- **Responsibilities:** Oversee hospital-wide operations; approve schedule and billing exceptions; monitor performance through dashboards; approve staff role assignments; ensure operational compliance.
- **Key Permissions:** View access across all clinical and operational modules; Approval authority over schedules, billing exceptions, and insurance escalations; Full management of reporting dashboards; Approval authority (not creation) over user role assignments.
- **Modules Accessible:** All modules (primarily View and Approve level); Reports & Dashboards (full management); User Management (approval only); Audit Logs (view only).

10. System Administrator

- **Responsibilities:** Manage user accounts, roles, and system-level access; configure system parameters; monitor audit logs for security and compliance; provide technical support for system functionality.
- **Key Permissions:** Full management of user accounts and role assignments; Full access to audit logs; No access to clinical, financial, or patient data (technical role only, per least-privilege principle).
- **Modules Accessible:** User Management, Audit Logs, Reports & Dashboards (system usage statistics only).

Governing Principles

- **Least Privilege:** Each role is granted the minimum access required to perform its function; no role has blanket access across all modules except where oversight explicitly requires it (Hospital Administrator, Reports; System Administrator, User Management/Audit Logs).
- **Segregation of Duties:** Clinical roles (Doctor, Nurse) are separated from financial roles (Billing Executive, Insurance Coordinator) to prevent conflicts of interest and reduce fraud risk.
- **Data Ownership Boundary:** Clinical staff can only access records for patients under their direct care ("assigned" scope), not the full patient database.
- **Administrative Separation:** Business oversight (Hospital Administrator) is kept distinct from technical system control (System Administrator), ensuring no single role holds both operational and infrastructural authority.

12.Functional Requirements

ID	Requirement
FR-1	The system shall allow new patients to register by entering mandatory details including name, date of birth, gender, contact number, email address, and address
FR-2	The system shall auto-generate a unique Patient ID upon successful registration
FR-3	The system shall allow patients to log in using registered email and password with secure authentication
FR-4	The system shall allow patients to reset their password via OTP sent to registered email or mobile number

FR-5	The system shall allow authorized staff to view, update, and manage patient profile information
FR-6	The system shall allow patients to view real-time doctor availability by department, date, and time slot
FR-7	The system shall allow patients to book appointments by selecting department, doctor, date, and available time slot
FR-8	The system shall prevent double booking by blocking already reserved time slots from selection
FR-9	The system shall auto-generate a unique Appointment ID upon successful booking
FR-10	The system shall allow patients to cancel or reschedule upcoming appointments with mandatory reason selection
FR-11	The system shall send automated SMS and email confirmation upon appointment booking, cancellation, and rescheduling
FR-12	The system shall send automated appointment reminder notifications to patients 24 hours before the scheduled appointment
FR-13	The system shall allow hospital administrators to create, update, and manage doctor profiles and schedules
FR-14	The system shall allow administrators to block doctor availability for leave, holidays, or unavailability
FR-15	The system shall allow doctors to view their daily and weekly appointment schedule upon login
FR-16	The system shall allow doctors to review complete patient medical history during consultation
FR-17	The system shall allow doctors to update diagnosis notes and generate digital prescriptions
FR-18	The system shall maintain a centralized electronic medical record for each patient including medical history, diagnoses, prescriptions, and visit records
FR-19	The system shall allow only authorized users such as doctors and administrators to update patient medical records
FR-20	The system shall record timestamp and user details for every update made to a patient record
FR-21	The system shall allow doctors to generate and store discharge summaries upon patient discharge

FR-22	The system shall allow doctors to raise laboratory test requests directly from the patient record
FR-23	The system shall allow lab technicians to upload completed test reports against the respective patient record
FR-24	The system shall notify patients via SMS or email when laboratory reports are available
FR-25	The system shall allow patients and doctors to view laboratory reports from the patient portal
FR-26	The system shall allow pharmacists to view digital prescriptions issued by doctors
FR-27	The system shall allow pharmacists to update medication dispensing records against each prescription
FR-28	The system shall maintain a complete prescription and dispensing history for each patient
FR-29	The system shall automatically generate itemized invoices based on consultation, tests, procedures, and medications
FR-30	The system shall support multiple payment modes including cash, card, and online payment
FR-31	The system shall allow billing executives to process and record patient payments
FR-32	The system shall allow insurance coordinators to submit and track insurance claims against patient bills
FR-33	The system shall update claim status in real time upon approval or rejection by the insurance provider
FR-34	The system shall provide role-based access control restricting system features based on user role including patient, doctor, receptionist, administrator, pharmacist, lab technician, and billing executive
FR-35	The system shall allow system administrators to create, modify, and deactivate user accounts and assign roles
FR-36	The system shall prevent unauthorized users from accessing restricted modules or patient data
FR-37	The system shall provide an administrative dashboard displaying real-time KPIs including daily appointments, patient count, revenue, and bed occupancy
FR-38	The system shall allow administrators to generate and export operational reports in PDF and Excel format

FR-39	The system shall maintain a complete audit trail logging all key system activities including login, data access, updates, and deletions with timestamp and user details
FR-40	The system shall allow administrators to monitor system usage and user activity logs
FR-41	The system shall allow patients to download and print invoices and lab reports from the patient portal
FR-42	The system shall automatically log out inactive users after a defined session timeout period

Functional Modules

Appointment Management

- Book, reschedule, and cancel appointments
- View doctor availability
- Receive appointment reminders and notifications

Patient Registration & Medical Records

- Patient onboarding and profile creation
- Electronic Medical Records (EMR) management
- Medical history and visit tracking

Doctor Management

- Doctor schedule management
- Consultation and prescription management
- Patient history review

Laboratory Management

- Test request and report upload
- Patient report tracking

Pharmacy Management

- Prescription management
- Medication dispensing records

Billing & Insurance

- Invoice generation
- Payment processing
- Insurance claim management

Reporting & Administration

- Dashboard and performance reports
- User and role management
- Audit logs and system monitoring

13.Non-Functional Requirements

ID	Requirement
NFR-01	The system shall respond to user requests within 3 seconds under normal operating conditions.
NFR-02	The system shall maintain a minimum uptime of 99.5% excluding scheduled maintenance.
NFR-03	The system shall support concurrent access by multiple users across departments.
NFR-04	The system shall be scalable to accommodate future expansion to additional hospital branches and departments.
NFR-05	The system shall enforce role-based access control to ensure authorized access to patient data.
NFR-06	The system shall encrypt sensitive patient and payment data during transmission and storage.
NFR-07	The system shall provide an intuitive and user-friendly interface for patients and hospital staff.
NFR-08	The system shall maintain audit logs of critical user activities and system events.
NFR-09	The system shall perform regular data backups and support disaster recovery procedures.
NFR-10	The system shall comply with applicable healthcare data privacy and security regulations.

NFR-11	The system shall be compatible with major browsers including Chrome, Firefox, Edge, and Safari
NFR-12	The system shall display clear and meaningful error messages to guide users on corrective actions
NFR-13	The system shall support data export in PDF and Excel formats for reports and summaries

Non-Functional Requirement Summary

The system must provide high performance, security, reliability, scalability, and usability to support hospital operations effectively. Strong access controls, data encryption, audit logging, backup mechanisms, and regulatory compliance are essential to protect sensitive patient information and ensure uninterrupted service.

14.Acceptance Criteria

User Story – US – 01: Patient Registration

AC-ID	Criteria	Expected Result
AC-01	Patient must fill all mandatory fields (Name, DOB, Gender, Contact, Address)	System highlights empty mandatory fields and prevents submission
AC-02	System must validate email format and mobile number	Error message displayed for invalid email/phone format
AC-03	Duplicate patient registration must be prevented	System detects existing record and shows "Patient Already Registered" alert
AC-04	Unique Patient ID must be auto-generated upon successful registration	Patient ID assigned and displayed on confirmation screen
AC-05	Patient receives confirmation via SMS/email after registration	Confirmation message delivered within 2 minutes of registration

User story – US-02- Booking Appointments

AC-ID	Criteria	Expected Result
AC-01	Patient must select Department, Doctor, Date, and Time slot	All fields mandatory; system prevents booking without complete selection
AC-02	Only available time slots must be displayed	Booked or blocked slots are greyed out and not selectable

AC-03	System must prevent double booking for same doctor at same time	System displays error message — Slot Already Booked — and prompts patient to select another available slot.
AC-04	Appointment confirmation with unique Appointment ID must be generated	Confirmation screen shows Appointment ID, Doctor, Date, and Time
AC-05	Patient receives SMS/email confirmation after booking	Notification delivered within 2 minutes with appointment details
AC-06	Automated reminder must be sent 24 hours before appointment	Reminder SMS/email triggered automatically 24 hours prior
AC-07	Patient cannot book appointment for a past date	System disables past dates on calendar and shows validation message

User story- US-04: Cancel / Reschedule Appointment

AC-ID	Criteria	Expected Result
AC-01	Patient can cancel or reschedule only upcoming appointments	Past appointments are non-editable and displayed as read-only
AC-02	Cancellation must require a reason selection before confirming	System prompts reason dropdown; submission blocked without selection
AC-03	Cancelled slot must become immediately available for other patients	Slot reflects as available in real time upon cancellation
AC-04	Patient receives SMS/email confirmation on cancellation or reschedule	Notification sent within 2 minutes with updated appointment details
AC-05	Rescheduled appointment must generate a new Appointment ID	New Appointment ID issued and old record marked as rescheduled

US-05: Manage Appointments (Admin/Doctor View)

AC-ID	Criteria	Expected Result
AC-01	Admin/Doctor can view all appointments filtered by date, department, or status	Filtered results displayed accurately within 3 seconds
AC-02	Admin can manually update appointment status (Confirmed, Completed, No-Show)	Status updated in real time and reflected across all views
AC-03	Doctor can view daily appointment schedule on login dashboard	Dashboard displays today's appointments in chronological order

AC-04	Admin can block doctor availability for leave or holiday	Blocked slots reflect immediately; no new bookings allowed for that period
AC-05	System must generate daily appointment summary report	Report downloadable in PDF/Excel format with accurate appointment count

US-08: Update Patient Records

AC-ID	Criteria	Expected Result
AC-01	Only authorized users (Doctor, Admin) can update patient records	Unauthorized roles receive "Access Denied" message on attempt
AC-02	All updates must be saved with timestamp and updated-by user details	Audit log entry created for every record modification
AC-03	Patient medical history, diagnosis, and prescriptions must be updatable	Changes saved successfully and reflected immediately on patient profile
AC-04	System must retain previous record versions for audit purposes	Previous versions accessible via record history/audit trail
AC-05	Mandatory fields cannot be left blank during record update	System highlights blank mandatory fields and blocks save action

15.Success Criteria

Functional Success Criteria

1. **Appointment Booking** — Patients can successfully book, reschedule, and cancel appointments online within 3 clicks.
2. **Patient Records** — 100% of patient data is digitized, centralized, and retrievable within 5 seconds.
3. **Automated Reminders** — Appointment reminders sent to patients via SMS/email at least 24 hours in advance.
4. **Billing Accuracy** — System generates error-free bills with automated insurance claim submission.
5. **Role-Based Access** — Every user role (doctor, nurse, admin, patient) has correct access permissions with zero unauthorized access.

Operational Success Criteria

6. **Wait Time Reduction** — Average patient wait time reduced by at least **40%** post-implementation.
7. **No-Show Reduction** — Appointment no-show rate reduced by at least **30%** through automated reminders.
8. **Staff Productivity** — Administrative manual workload reduced by at least **50%** through automation.
9. **System Uptime** — System maintains minimum **99%** uptime during operational hours.
10. **Data Migration** — 100% of existing patient records successfully migrated with no data loss.

Technical Success Criteria

11. **Performance** — System handles peak load of concurrent users without performance degradation.

12. **Security** — Zero data breaches reported within first 6 months of go-live.

13. **Integration** — Successful real-time integration with lab, pharmacy, and insurance systems.

14. **Scalability** — System supports addition of new branches/departments without major rework.

User Satisfaction Criteria

15. **User Adoption** — Minimum 85% of staff actively using the system within 30 days of go-live.

15\. **Patient Satisfaction** — Patient satisfaction score improves by minimum 30% in post-launch survey.

17. **Training Completion** — 100% of end-users complete system training before go-live.